

SPESIFIKASI ARSITEKTUR FRONTEND & UI/UX DESIGN SYSTEM

SISTEM PENGURUSAN R1/R2/R3 MIMOS ACADEMY

Versi Dokumen: 1.0.0

Status: Enterprise Frontend Blueprint

Peranan Pasukan: Chief Product Officer | UX Director | Enterprise Architect | Lead Frontend Engineer

1. STRATEGI REKA BENTUK & VISI PRODUK

1.1 Persona & Estetika Reka Bentuk

Aplikasi ini direka khas untuk peringkat **Executive & Corporate Enterprise** di MIMOS Academy. Rekabentuk mengadunkan elemen terbaik daripada:

- **Microsoft Power BI & Fabric:** Untuk kepadatan data (data density), pemformatan nombor yang tepat, dan KPI Cards yang berasaskan maklumat eksekutif.
- **Salesforce CRM:** Untuk struktur penjejakan rantai perniagaan (*Opportunity-to-Cash-to-Delivery*), status pautan bertingkat, dan *Program 360 View*.
- **HubSpot:** Untuk kejelasan *Action Center*, kebersihan *user flow*, dan pengurusan status tindakan (*Task & Follow-up*).
- **Modern SaaS Dashboard:** Kebersihan susun atur, tipografi yang jelas, *micro-interactions*, dan ruang positif (*white space*) yang seimbang.

Pendekatan Estetika: *Corporate / Executive / High-Trust Enterprise UI.*

BUKAN Style Startup (elakkan gradien neon), **BUKAN** Style Gaming (elakkan mod gelap cyberpunk), dan **BUKAN** Colourful Style (elakkan warna gula-gula/pastel tidak profesional).

1.2 Palet Warna & Panduan Gaya (UI Style Guide)

Warna Utama (Brand & Surface)

- **Primary Deep Navy:** #0F172A (Slate 900) – Header, Sidebar Utama, Teks Utama, Modal Header.
- **Brand Royal Blue:** #2563EB (Blue 600) – Aksyen Utama, Active State, Primary Buttons, Active Tabs.
- **Hover Royal Blue:** #1D4ED8 (Blue 700) – State hover butang utama.
- **Canvas Background:** #F8FAFC (Slate 50) – Latar belakang keseluruhan dashboard.
- **Card / Surface White:** #FFFFFF – Latar belakang modul, kad, dan jadual.

- **Subtle Border / Divider:** #E2E8F0 (Slate 200) – Garisan pemisah modul dan grid jadual.
- **Neutral Secondary Text:** #64748B (Slate 500) – Sub-tajuk, label borang, dan *secondary metrics*.

Warna Semantik (Status & Indicator)

- **Success Green:** #059669 (Emerald 600) – Paid, Approved, Secured, Completed.
- **Success Surface:** #ECFDF5 (Emerald 50) – Latar belakang badge status *Paid*.
- **Warning Amber:** #D97706 (Amber 600) – Pending, In Progress, Follow-up Required, Overdue < 30 hari.
- **Warning Surface:** #FFFBEF (Amber 50) – Latar belakang badge status *Pending*.
- **Danger Red:** #DC2626 (Red 600) – Overdue > 30 hari, Rejected, Missing Document, Unmatched Record.
- **Danger Surface:** #FEF2F2 (Red 50) – Latar belakang badge *Critical Alert*.
- **Info Blue / Cyan:** #0284C7 (Sky 600) – Quotation Sent, Forecast Stage, Info System.

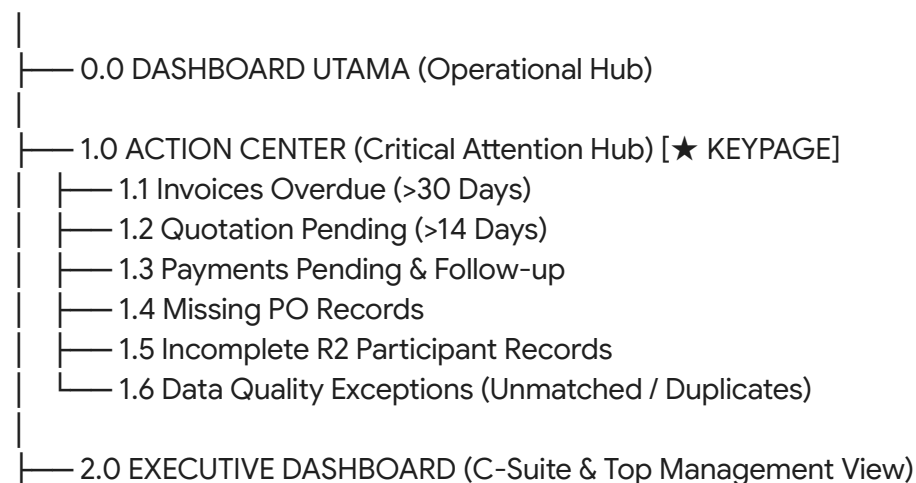
1.3 Tipografi & Skala Hirarki

Gunakan muka taip **Inter** atau **Plus Jakarta Sans** untuk keterbacaan tinggi pada skrin berdensiti data:

- **Display / CEO Metrics:** 32px | SemiBold | Line Height 1.2 (Nilai Ringgit KPI Utama)
- **Heading 1 (Page Title):** 24px | Bold | Line Height 1.3
- **Heading 2 (Section Title):** 18px | SemiBold | Line Height 1.4
- **Heading 3 (Card Title):** 15px | Medium | Line Height 1.4
- **Body Text Standard:** 14px | Regular | Line Height 1.5
- **Table Data / Dense Specs:** 13px | Regular | Line Height 1.4
- **Captions & Badges:** 11px | SemiBold | Upper/Normal Case

2. INFORMATION ARCHITECTURE (IA) & SITEMAP LENGKAP

SISTEM PENGURUSAN R1/R2/R3 MIMOS ACADEMY



—	2.1 Executive Revenue & Cashflow Summary
—	2.2 Conversion Pipeline & Funnel Performance
—	2.3 R2 Training Impact & Demographic Trends
—	2.4 High-Value Opportunity Risk Matrix
—	3.0 R3 SALES & OPPORTUNITY FUNNEL
—	3.1 All Opportunities (Pipeline Master List)
—	3.2 Lead Tracking
—	3.3 Sales Forecast (Weighted Revenue)
—	3.4 Pipeline Kanban / Stage Board
—	4.0 QUOTATION MANAGEMENT
—	4.1 Quotation Master List
—	4.2 Pending Approval / Sent
—	4.3 Approved Quotations
—	5.0 PURCHASE ORDERS (PO)
—	5.1 PO Monitoring Hub
—	5.2 PO vs Quotation Matching Status
—	5.3 Missing PO Escalation Queue
—	6.0 R1 FINANCIALS (INVOICES & PAYMENTS)
—	6.1 Invoices Master List
—	6.2 Outstanding Invoices & Collection Aging (30/60/90+ Days)
—	6.3 Collections & Payment Received
—	6.4 Payment Follow-Up Queue
—	6.5 Cost of Sales & Profitability Analysis
—	7.0 R2 TRAINING MANAGEMENT
—	7.1 Program Directory (Master Program Catalog)
—	7.2 Training Session Calendar & Schedule
—	7.3 Participant Demographics (Bumiputera vs Non-Bumiputera)
—	7.4 Training Statistics & Impact Metrics
—	8.0 PROGRAM 360° VIEW (Unified Single Program CRM Page) [★ KEYPAGE]
—	8.1 Program Overview & Client Profile
—	8.2 End-to-End Chain Tracker (Lead → PO → Invoice → Training)
—	8.3 Financials & Payment Breakdown
—	8.4 Delivery & Participant Roster
—	8.5 Audit & Activity History
—	9.0 REPORTS & ANALYTICS
—	9.1 R1 Income & Financial Statement Report

—	9.2 R2 Overall Training Report
—	9.3 R3 Sales Funnel Report
—	9.4 Executive Board Pack Export Generator
—	10.0 DATA MANAGEMENT & QUALITY CENTER
—	10.1 Import Center (Excel Drag & Drop Parser)
—	10.2 Import Logs & Validation History
—	10.3 Data Mismatch & Reconciliations
—	10.4 Duplicate Detection & Resolution
—	11.0 SYSTEM GOVERNANCE & SETTINGS
—	11.1 User & Role Access Management (RBAC)
—	11.2 Master Data Dictionary (Client, Account Aliases)
—	11.3 Status & Pipeline Stage Mapping Rules

3. STRUKTUR NAVIGASI (NAVIGATION STRUCTURE)

Sistem menggunakan **Dual-Level Navigation Strategy**:

1. Top Navigation Bar (Fixed 64px):

- **Kiri:** MIMOS Academy Logo + System Switcher ("R1/R2/R3 Executive Portal").
- **Tengah:** Global Search Bar (Cmd + K / Ctrl + K) – Boleh cari Program, Quotation No, Invoice No, Client Name.
- **Kanan:** Quick Action Button (" + New Record" / "Upload Excel"), Notification Bell (dengan jumlah alert Action Required), User Profile Dropdown dengan Peranan (e.g. *Super Admin / PIC / Management*).

2. Left Sidebar Navigation (Collapsible - 260px expanded / 72px collapsed):

Kumpulan Navigation	Menu Item	Icon (Lucide React)	Peranan Akses (RBAC)
Utama	Dashboard Utama	LayoutDashboard	Semua Peranan
	Action Center	AlertTriangle (Badge Red)	Admin, PIC, Manager
	Executive Board	TrendingUp	Management, Manager, Admin
R3 (Jualan)	R3 Sales Funnel	Filter	Admin, PIC, Manager

	Quotations	FileText	Admin, PIC, Manager
	Purchase Orders	FileCheck	Admin, PIC, Manager
R1 (Kewangan)	Invoices & Aging	Receipt	Admin, PIC, Manager
	Payments Received	CreditCard	Admin, PIC, Manager
R2 (Latihan)	Training Programs	GraduationCap	Admin, PIC, Manager
	Participant Analytics	Users	Admin, PIC, Manager, Viewer
Tadbir Data	Import Center	UploadCloud	Admin, PIC
	Data Quality Audit	ShieldAlert	Admin, Manager
	Reports Hub	BarChart3	Semua Peranan
Sistem	Users & Roles	UserCog	Admin
	Settings & Mapping	Sliders	Admin

4. USER FLOW & RANTAIAN END-TO-END DATA (USER JOURNEY)

4.1 Chain Architecture

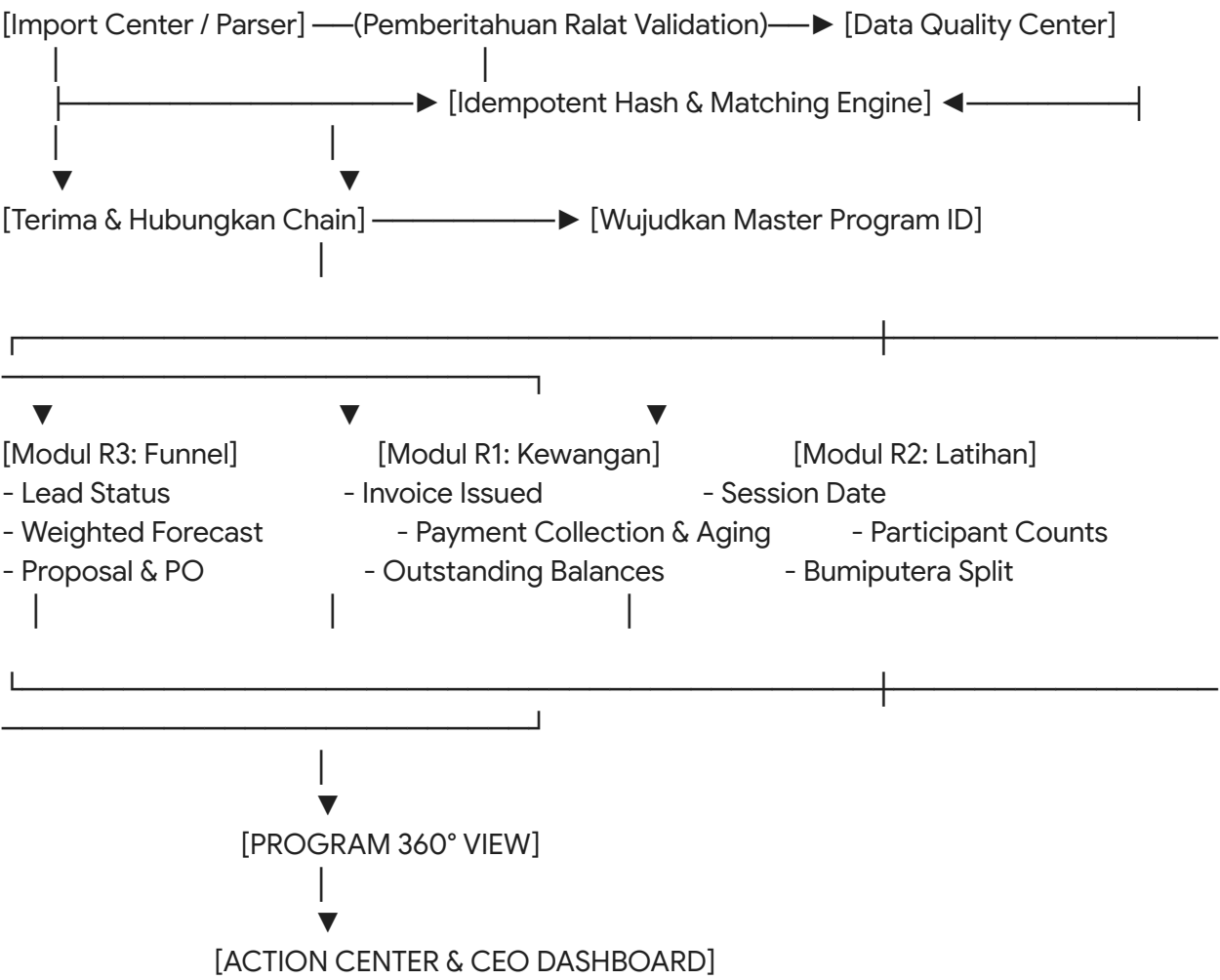
Sistem memastikan setiap rekod terhubung secara konsisten:

Lead (R3) → Opportunity → Quotation → PO → Invoice (R1) → Payment → Training (R2)

4.2 Primary User Flow Diagram

[Mula: Excel Import / Manual Input]

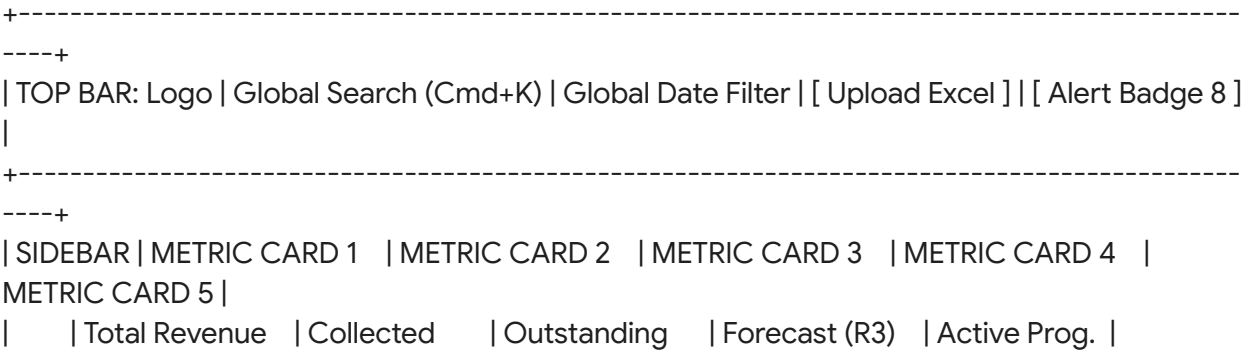




5. REKA BENTUK SKRIN UTAMA (SCREEN SPECIFICATIONS)

SKRIN 1: DASHBOARD UTAMA (OPERATIONAL HUB)

Wireframe & Layout Architecture



	RM 3,450,000	RM 2,100,000	RM 1,350,000	RM 1,820,000	42 Sessions	
	+-----+					
	SECTION LEFT (65% Width)		SECTION RIGHT (35% Width)			
	+-----+		+-----+			
	VISUAL 1: Revenue vs Collection Trend		ACTION REQUIRED TODAY (Top 5)			
	(Bar + Line Combo Chart: Monthly)		→ Inv #1042 Overdue 45 days (RM24k)			
	+-----+		→ Quo #0088 Pending >14 days			
	VISUAL 2: Sales Funnel Breakdown (R3)		→ Prog #AI-02 Missing R2 Data			
	(Funnel Stage Horizontal Progress Bars)		+-----+			
	+-----+		VISUAL 3: Upcoming Training (R2)			
	TABLE: Recent Chain Activities & Updates		List of next 7 days sessions			
	+-----+					

Komponen Skrin

1. Top KPI Grid (5 Cards):

- *Total Revenue (Secured & Invoiced)* | RM | Micro-sparkline trend
- *Collected Revenue (R1)* | RM | Status indicator (% daripada Invoiced)
- *Outstanding Collection (R1)* | RM | Red indicator jika overdue > 30 hari
- *Forecast Weighted Revenue (R3)* | RM | Unjuran jualan berasaskan kebarangkalian
- *Total Participants Trained (R2)* | Count | Pecahan Bumi vs Non-Bumi micro bar

2. Main Revenue & Collection Chart:

- Carta Gabungan: Bar (Invoiced Amount) vs Line (Cash Collected) mengikut bulan.

3. Sales Funnel Summary Bar:

- Bar peringkat: Lead → Proposal → Negotiation → PO Signed → Invoiced.

4. Action Required Widget (Direct Link ke Action Center):

- 5 item keutamaan tertinggi dengan butang *Quick Action* ("Follow-up", "View Chain").

SKRIN 2: ACTION CENTER ("ACTION REQUIRED") [★ HALAMAN PALING PENTING]

Halaman ini merupakan **Nadi Operasi** sistem untuk menghentikan pembocoran hasil (*revenue leakage*) dan memastikan keselarasan data.

Wireframe Layout

+	+-----+				
----	+				
	HEADER: Action Required Center [Filters: Status PIC Date] [Export Critical Summary]				
	Subtitle: 18 Perkara Memerlukan Tindakan Segera Hari Ini				
	+-----+				
----	+				

TAB SYSTEM:					
[ALL ALERTS (18)] [OVERDUE INVOICES (5)] [PENDING QUO (4)] [MISSING R2 (6)] [DATA ERR (3)]					
+-----+					
----+					
SEVERITY BANNER: 5 Critical Overdue Invoices Totaling RM 184,200.00					
+-----+					
----+					
TABLE / CARD LISTING OF ACTION ITEMS					
Priority Category Title / Client & Document Issue Detail PIC Action					
+-----+-----+-----+-----+-----+					

[HIGH] Invoice Overdue MINDEF - RM 46,285.00 Overdue 52 Days Fuzy					
[Resolve]					
Inv: MA/INV/2026/0012 Payment Status: UNPAID [Email]					
+-----+-----+-----+-----+-----+					

[HIGH] Missing Document TNB ILSAS - RM 28,000.00 Invoiced without PO Adila					
[Upload PO]					
+-----+-----+-----+-----+-----+					

[MED] Pending Quotation PUNB - RM 17,850.00 Pending >18 Days Suhairi					
[Reminder]					
+-----+-----+-----+-----+-----+					

[MED] Incomplete R2 Canva AI Cohort 2 Missing Participant Fuzi [Input R2]					
(Completed 3 days ago) Demographics Breakdown					
+-----+					
----+					

Logik Kategori Amaran (Action Rules Matrix)

- Invoices Overdue > 30 Hari:** Flag Merah. Paparkan hari outstanding, jumlah RM, dan butang cepat log *Follow-up Note*.
- Quotation Pending > 14 Hari:** Flag Jingga. Notifikasi kepada PIC untuk kemaskini status jualan (*Win/Lost/Follow-up*).
- Missing PO:** Flag Jingga. Invois telah terbit tetapi fail PO rasmi belum dimuat naik.
- Incomplete R2 Records:** Latihan telah tamat (berdasarkan tarikh session) tetapi data

bilangan peserta Bumi/Non-Bumi masih kosong.

5. **Unmatched / Duplicate Records:** Data Excel dimuat naik gagal menemui padanan *Quotation/Invoice* atau terdapat duplikasi borang.

SKRIN 3: PROGRAM 360° VIEW (CRM UNIFIED PAGE) [★ HALAMAN UNIFIKASI DATA]

Paparan tunggal (*Single View of Truth*) yang menggabungkan rantaian perniagaan penuh untuk sesuatu program dalam satu halaman.

Wireframe Architecture

+-----+			
+-----+			
TOP NAVIGATION & BACK BUTTON: ◀ Back to Programs Directory			
+-----+			
+-----+			
HEADER CARD: Certified AI Trainer & Integration (MINDEF Cohort 1)			
Client: Kementerian Pertahanan Malaysia (MINDEF) Category: Artificial Intelligence PIC: Fuzy			
Program ID: PRG-2026-0042 Global Status: INVOICED & DELIVERED			
+-----+			
+-----+			
VISUAL CHAIN PROGRESS TRACKER (Stepper Component)			
[✓] Lead Registered —▶ [✓] Quotation Approved —▶ [✓] PO Received —▶ [✓] Invoiced			
—▶ [!] Payment Pending —▶ [✓] Training Done			
12 Jan 2026 QT-2026-0011 PO-994821 INV-2026-004 (Overdue 12			
Days) 18 Feb 2026			
+-----+			
+-----+			
TABBED DETAILED SECTIONS:			
[OVERVIEW & CHAIN] [FINANCIALS & INVOICING] [R2 TRAINING & PARTICIPANTS] [			
AUDIT LOG]			
+-----+			
+-----+			
LEFT COLUMN (60% Width) RIGHT COLUMN (40% Width)			
+-----+			
+-----+			
FINANCIAL CHAIN SUMMARY CLIENT & CONTACT DETAILS			
Proposal Value : RM 50,000.00 Client Name: MINDEF Malaysia			
Approved Quo Value: RM 46,285.00 Contact Person: Major Azman			
PO Value : RM 46,285.00 Email: azman@mindef.gov.my			
Invoiced Amount : RM 46,285.00 (Inc 8% SST)			
+-----+			

```

|| Amount Collected : RM 0.00 (UNPAID)    || QUICK ACTIONS          || |
| +-----+ || [ Log Payment Received ]      ||
|| R2 PARTICIPANT BREAKDOWN                ||| [ Upload Attendance List ] ||
|| Total Participants: 25                   ||| [ Generate Invoice PDF ]   ||
|| Bumiputera      : 22 (88%)              || +-----+ |
|| Non-Bumiputera  : 3 (12%)              || ACTIVITY & FOLLOW-UP HISTORY ||
|| Venue           : MIMOS Computer Lab 2  || • 20 Feb: Email reminder sent to Major Azman ||
+-----+
--+

```

SKRIN 4: IMPORT CENTER (EXCEL INGESTION & RECONCILIATION)

Pusat pemprosesan muat naik fail Excel (R1, R2, R3, Quotation, Invoices) secara profesional.

Component Layout & Flow

1. Drag and Drop Zone:

- Sokongan fail: .xlsx, .xls, .csv.
- Panduan templat automatik (*Download Master Excel Template*).

2. Live Upload Progress & Validation Pipeline:

- Step 1: File Integrity Check → Step 2: Column Mapping Verification → Step 3:
Duplicate & Unmatched Record Detection → Step 4: Final Commit to System Database.

3. Import Audit Summary Widget:

- **Total Rows Parsed:** e.g. 142 rows.
- **New Records Created:** 12.
- **Existing Records Updated:** 125.
- **Rejected / Validation Errors:** 5 rows (dengan pautan *View Error Log*).

4. Error Reconciliation Table:

- Paparkan baris spesifik fail Excel yang rosak (cth: #REF!, #DIV/0!, tarikh tidak sah, nombor Quotation bercanggah) dengan butang *Inline Fix* sebelum import diluluskan.

SKRIN 5: DATA QUALITY & AUDIT CENTER

Skrin khas untuk mengaudit dan membersihkan *data debt* perniagaan.

Features & Widgets

1. Data Mismatch Cards:

- Contoh: Status bayaran dalam R1 memaparkan *PAID* tetapi fail *Invoice Tracker* memaparkan *UNPAID*.

2. Unmatched Client Entities:

- Pengurusan Alias Syarikat (Contoh: Menggabungkan "MIMOS Berhad", "MIMOS", dan

"MB" menjadi satu Master Account).

3. Duplicate Quotation & Invoice Detector:

- Senarai rekod dengan nombor sebut harga atau nilai yang bertindih untuk disemak oleh Admin.

SKRIN 6: EXECUTIVE DASHBOARD (CEO & TOP MANAGEMENT VIEW)

Reka bentuk berimpak tinggi bertaraf C-Level untuk memberi gambaran keseluruhan prestasi MIMOS Academy dalam tempoh 5 saat.

Layout & Visuals

- **Executive Metrics Strip:** Revenue Secured, Cash Collected, Gross Profit Margin %, Pipeline Coverage Ratio.
- **Waterfall Revenue Visual:** Daripada *R3 Sales Forecast* → *Secured PO* → *Invoiced Revenue* → *Net Cash Inflow*.
- **R2 Demographic Impact Bar:** Nisbah penyertaan Bumiputera vs Non-Bumiputera mengikut sektor industri.
- **Top Client Contribution Matrix:** 10 Klien Utama mengikut nilai sumbangan kewangan.

6. SYSTEM COMPONENT LIBRARY & DESIGN SYSTEM

6.1 Core UI Atoms & Molecules (React / Tailwind UI Components)

1. Executive Status Badge (<StatusBadge />)

// Variasi status kanonik dengan pematuhan warna semantik enterprise

```
type StatusType = 'PAID' | 'UNPAID' | 'OVERDUE' | 'PENDING' | 'INVOICED' | 'DRAFT';
```

```
interface StatusBadgeProps {  
  status: StatusType;  
  label?: string;  
}
```

```
export const StatusBadge: React.FC<StatusBadgeProps> = ({ status, label }) => {  
  const styles = {  
    PAID: 'bg-emerald-50 text-emerald-700 border-emerald-200',  
    UNPAID: 'bg-amber-50 text-amber-700 border-amber-200',  
    OVERDUE: 'bg-rose-50 text-rose-700 border-rose-200 animate-pulse',  
    PENDING: 'bg-slate-100 text-slate-700 border-slate-200',  
    INVOICED: 'bg-sky-50 text-sky-700 border-sky-200',  
    DRAFT: 'bg-gray-50 text-gray-600 border-gray-200',  
  };  
};
```

```
return (
```

```

    <span className={`inline-flex items-center px-2.5 py-0.5 rounded-full text-xs font-semibold
border ${styles[status]}`} >
      <span className="w-1.5 h-1.5 rounded-full fill-current mr-1.5 shrink-0" />
      {label || status}
    </span>
  );
};

```

2. Visual Chain Stepper (<ChainStepper />)

Menampilkan kemajuan rantai perniagaan secara visual daripada Lead ke Latihan:

```

interface ChainStep {
  id: string;
  label: string;
  date?: string;
  status: 'completed' | 'current' | 'upcoming' | 'error';
}

export const ChainStepper: React.FC<{ steps: ChainStep[] }> = ({ steps }) => {
  return (
    <div className="flex items-center justify-between w-full py-4 px-6 bg-white rounded-xl
border border-slate-200 shadow-sm">
      {steps.map((step, idx) => (
        <React.Fragment key={step.id}>
        <div className="flex items-center space-x-3">
          <div className={`w-8 h-8 rounded-full flex items-center justify-center text-xs font-bold
${
            step.status === 'completed' ? 'bg-blue-600 text-white' :
            step.status === 'current' ? 'bg-amber-500 text-white ring-4 ring-amber-100' :
            step.status === 'error' ? 'bg-rose-600 text-white' : 'bg-slate-100 text-slate-400'
          }`} >
            {step.status === 'completed' ? '✓' : idx + 1}
          </div>
          <div>
            <p className="text-xs font-semibold text-slate-900">{step.label}</p>
            {step.date && <p className="text-[11px] text-slate-500">{step.date}</p>}
          </div>
        </div>
        {idx < steps.length - 1 && (
          <div className={`flex-1 h-0.5 mx-4 ${step.status === 'completed' ? 'bg-blue-600' :
'bg-slate-200'}`} />
        )}
      )}
    </div>
  );
};

```

```

    </React.Fragment>
  )}
</div>
);
};

```

3. Enterprise Data Table Header (<DataTableHeader />)

Diwujudkan khas untuk jadual berdensiti data tinggi dengan sokongan penyaringan ringkas.

7. CADANGAN SENI BINA FRONTEND REACT (FRONTEND ARCHITECTURE)

7.1 Tech Stack Standard

- **Framework:** React 18+ (dengan Vite / Next.js SPA Mode).
- **Language:** TypeScript (Strict Mode untuk ketepatan data kewangan).
- **Styling:** Tailwind CSS (dengan pemuat warna tema *Deep Navy* dan *Royal Blue*).
- **Icons:** Lucide React (lucide-react).
- **Data Table:** TanStack Table v8 (React Table) – Untuk sorting, filtering, server-side pagination, dan virtualization.
- **Charts & Analytics:** Recharts / Apache ECharts React – Untuk graf kewangan eksekutif.
- **State Management:** Zustand (Client UI State & Active Filters) + TanStack Query / React Query (Mock Data Hydration & Caching).
- **Form & Validation:** React Hook Form + Zod (Validation schema untuk borang).

7.2 Directory & Folder Structure (Clean Feature-Based Architecture)

```

src/
├── assets/           # Logos, Static Images, Fonts
├── components/       # Shared UI Components (Design System)
│   ├── ui/          # Buttons, Inputs, Modals, Badges, Cards
│   ├── feedback/     # Toast, Alert Banners, Loading Spinners
│   ├── layout/       # Sidebar, Topbar, AppLayout, PageContainer
│   └── tables/       # EnterpriseDataTable, TablePagination, FilterBar
├── features/         # Feature Modules (Domain-Driven)
│   ├── dashboard/    # Operational Dashboard Components & Views
│   ├── action-center/ # Action Required Hub Components & Rules Engine
│   ├── program-360/  # Program 360 Unified Page Components
│   ├── r1-financials/ # Invoices, Payments, Aging Tables
│   ├── r2-training/  # Programs, Sessions, Participants Demographics
│   └── r3-funnel/    # Leads, Opportunities, Pipeline Board

```

— import-center/	# Excel Parser, Drag&Drop Dropzone, Error Fixer
— data-quality/	# Audit Dashboard, Duplicates & Mismatch Rules
— executive/	# CEO Dashboard & Board Visuals
— hooks/	# Custom React Hooks (e.g., useFilter, useDebounce)
— layouts/	# Main Dashboard Layout, Auth Layout
— mock-data/	# JSON Mocks (Extracted from MIMOS Excel Files)
— routes/	# React Router Config & Role-based Protection
— stores/	# Zustand Global Stores (FilterStore, UserStore)
— types/	# TypeScript Interfaces (Program, Invoice, R2Data)
— utils/	# Excel Parsers, Currency Formatters, Date Utilities

7.3 Model Data Types (TypeScript Definitions)

// Core Rantaian Data MIMOS Academy

```
export type PipelineStage =
  | 'LEAD_REGISTERED'
  | 'PROPOSAL_SUBMITTED'
  | 'QUOTATION_APPROVED'
  | 'PO_RECEIVED'
  | 'INVOICED'
  | 'PAID'
  | 'TRAINING_COMPLETED';
```

```
export interface Program360Record {
  programId: string;
  programTitle: string;
  clientName: string;
  clientCategory: 'GOVERNMENT' | 'CORPORATE' | 'INTERNAL' | 'FOC';
  picName: string;
  currentStage: PipelineStage;
```

// R3 Funnel Data

```
leadDate: string;
forecastValue: number;
probability: number; // e.g. 0.85
weightedValue: number;
```

// Quotation & PO Data

```
quotationNo?: string;
quotationDate?: string;
poNo?: string;
```

```

poAmount?: number;

// R1 Financial Data
invoiceNo?: string;
invoiceDate?: string;
invoiceAmount?: number;
sstAmount?: number;
amountCollected: number;
outstandingBalance: number;
daysOutstanding: number;
paymentStatus: 'PAID' | 'UNPAID' | 'PARTIAL' | 'OVERDUE';

// R2 Training Data
trainingStartDate?: string;
trainingEndDate?: string;
totalParticipants: number;
bumiputeraCount: number;
nonBumiputeraCount: number;
r2Status: 'COMPLETED' | 'PENDING_DATA' | 'UPCOMING';

// Action Required Flags
hasActionRequired: boolean;
actionRequiredReason?: string[];
}

```

8. STRATEGI RESPONSIF (RESPONSIVE LAYOUT MATRIX)

Sistem ini dioptimumkan untuk penggunaan merentas peranti dengan tumpuan khas:

Peranti	Grid Layout	Penyesuaian UI & UX
Desktop / Monitor (1440px+)	Full Grid Layout (Dual Column, 5 KPI Cards Row)	Pengalaman Penuh Enterprise: Jadual data berdensiti tinggi, graf bersebelahan, <i>Program 360 View</i> penuh.
Laptop (1024px - 1439px)	Flexible Grid (4 KPI Cards Row, Stacked Sidebars)	Sidebar dikuncupkan secara automatik kepada mod ikon sahaja untuk

		memaksimalkan ruang jadual.
Tablet (768px - 1023px)	2 Column Grid Layout	Jadual menggunakan <i>horizontal scroll container</i> . Graf dipaparkan secara bertingkat (<i>stacked</i>).
Mobile (320px - 767px)	Single Column Layout	Fokus Mobile Khusus: Dashboard KPI Ringkasan, Action Center Alert List , Program Details Check , dan Follow-up Note Logger . Navigation bertukar ke <i>Mobile Bottom Bar</i> .

9. DASHBOARD CADANGAN UNTUK IMPRESS CEO & TOP MANAGEMENT

Untuk memastikan papan pemuka (*dashboard*) memikat hati C-Level MIMOS Academy, 5 elemen impak tinggi berikut wajib disertakan pada antara muka utama:

- Metric Cash Conversion Speedometer:**
 - Menunjukkan kadar kepantasan kutipan wang daripada invois dikeluarkan hingga bayaran diterima dalam bentuk *Radial Gauge Visual*.
- Executive Waterfall Revenue Stream:**
 - Graf aliran wang yang jelas: *Total Forecast (R3)* → *Secured Contract/PO* → *Invoiced Revenue (R1)* → *Net Cash Collected*. CEO boleh melihat tempat berlakunya kebocoran hasil secara serta-merta.
- Real-time Revenue Leakage Radar:**
 - Widget bernilai tinggi yang menunjukkan jumlah RM ringgit yang berisiko terjejas disebabkan invois overdue >30 hari dan sebut harga menggantung >14 hari.
- R2 National Talent Impact Gauge:**
 - Visualisasi menarik pecahan *Bumiputera vs Non-Bumiputera* beserta jumlah modal insan yang telah dilatih oleh MIMOS Academy dalam tahun semasa.
- One-Click "Board Executive Report" Generator:**
 - Butang khas pada bahagian atas skrin yang membolehkan CEO mencetak atau mengeksport keseluruhan paparan ringkasan ke format PDF mesra cetakan mesyuarat lembaga pengarah secara automatik.

Disediakan oleh: Pasukan Rekabentuk & Arkitek Perisian Frontend MIMOS Academy System.